NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L26.7 Probing AI Friendliness of STAR-CCM+ vs. Fluent

Group Rapid Trials Lab

Status Promoted

Period 2026-10-19 19:10 — 2026-10-25 10:12

Stage conclusion

Project report

L26.7 STAR-CCM+ vs. Fluent: AI-friendliness probe

Criteria (written 10-19 19:11, not changed afterward; any change must be logged as a decision, with the old reading kept alongside)

Complete and provide a "Worth using" directional recommendation = both conditions met simultaneously: 1. ≥6 of the 7 comparison dimensions have conclusions backed by **official sources** (a dimension counts as "backed by official sources" only if both STAR and Fluent sides have official sources) 2. Clear judgments are given for both usage modes A and B: STAR is **more friendly / comparable / less friendly** than Fluent.

Otherwise judge as **Abandon** (specify which dimensions lack official sources and why they could not be found). Dimensions with insufficient sources must be marked "Not found" rather than padded with secondary articles.

7 dimensions: (1) Whether the API uniformly covers the full workflow (CAD→mesh→physics→solve→post-processing); (2) Language and LLM familiarity (STAR Java macros vs. Python/PyFluent settings objects); (3) GUI extension mechanism (STAR Java plugins vs. Fluent Scheme/UDF/ACT); (4) Volume of documentation and public examples; (5) Remote/HPC batch processing; (6) Official AI integration (MCP, vendor AI assistants); (7) Licensing and availability (including whether ND holds a license).

Operational definitions of criterion terms: - **Official source**: Pages published by Siemens (docs.sw.siemens.com, plm.sw.siemens.com, official Siemens blogs/press releases, official Siemens GitHub organizations) or Ansys/Synopsys (ansys.com, ansyshelp, fluent.docs.pyansys.com, github.com/ansys, official press releases); for ND licensing, only official ND sites (nd.edu domain) count. Secondary articles (including the original article that prompted this probe, forums, blogs, and third-party tutorials) serve only as leads and do not count as official sources. - **More friendly**: Under the given usage mode, STAR has an official capability in this dimension that Fluent lacks and that makes a substantive difference for building agents/assistants, without Fluent having an equivalent reverse advantage; **Comparable**: both sides have viable official paths, differing only in style or convenience; **Less friendly**: the converse. Overall judgment for a usage mode = majority direction across the dimensions relevant to that usage mode (A: (1)(2)(3)(6)(7); B: (1)(2)(4)(5)(6)(7)), except that if any dimension has a hard gap ("cannot do / non-existent"), the hard gap governs and must be stated explicitly. - **Countable volume of public examples**: State the exact counting methodology (e.g., number of repositories matching a GitHub keyword search, number of entries in the official example library, query date); do not mix different methodologies in comparisons. - **LLM familiarity**: Proxied solely by countable public code/corpus volume (e.g., GitHub code search hit counts, number of official examples), without live model benchmarking (this quick trial does not execute model comparisons).

Boundaries

- Judgment-type conclusions are made by the lead directly, not delegated to Gemini; web searches are used for evidence gathering.
- The Platform Lead's initial assessment ((A) STAR is more friendly; (B) little difference, Fluent+PyFluent may be more convenient) is treated strictly as a hypothesis to be tested, not as a conclusion.
- For ND licensing, check only the public university software catalog without logging in or touching credentials; if not found, write "Requires confirmation from the PI". If found available, add a 5-task mini benchmark **design** (design criteria and tasks only; do not execute).
- For the actual state of PyFluent on our platform, read first (read-only), and only look further if insufficient.

Trials and results

Time	Action taken	Result
10-19 19:11	Finalized header and criteria	—
10-19 21:20	Retroactively submitted launch plan; started M1 official-source evidence collection (read-only web search)	Evidence logged, in progress
10-21 14:30	Wrote progress and plan report; deadline extended to 10-22 19:00	Report
10-24 21:56	Deadline passed, settled per criteria (the coordinating lead instructed that no new material be collected)	**Abandon per criteria**: 0/7 dimensions with official sources on both sides; neither A nor B judged
10-25 10:10	Supplemented official sources for both sides (read-only public web pages, 0 compute)	Evidence table (20 direct reads); comparison table; both usage modes A and B judged as "STAR is less friendly than Fluent"

Key figures